

Motor Gear Reduction

Levi Terry

UAT

RBT205: Mechanisms and Materials

Prof. Prater

11/16/25

Motor Gear Reduction

- **Wheel Diameter:** 59.8mm (5.98 cm)
- **Motor RPM:** 250@6V
- **Walking speed of 20–29-year-old:** 3 mph

(<https://www.medicalnewstoday.com/articles/average-walking-speed#average-speed-by-age>)

Math

Mile to cm: 160,934

$3 \text{ mi/hr} \times 160934 \text{ cm} / 1\text{mi} = 482,802 / 60 = 8,046.7 \text{ cm/min}$

5.98 cm tire

$8,046.7 / 5.98 = 1,345.60 \text{ rev/min}$

Required Gear Ratio: $1,345.6 / 250 = 5.3824$

Gear Teeth Chosen: 8, 12, 16, 32

Actual Gear Ratio: $(16/12) + (12/36) + (36/8) = 6.166666667$

Video

